**Higher National Diploma in Engineering (Electrical)**
**Fourth Year, First Semester Examination – 2020**
**EE 3204– Electrical Energy Management**

**Instructions for Candidates:**
Answer **Five (05)** questions.
Each question carries equal marks

No. of questions:06
No. of pages : 04
Time : **03 Hours**

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Q1.

(a) (i) What do you understand by the term 'Electrical Energy Management'? (03 Marks)

(ii) Why do you need the electrical energy management concept? (03 Marks)

(b) (i) How do you check the polarity of transformer practically? Explain briefly with Suitable diagrams. (03 Marks)

(ii) What are the two main tests carried out in single phase transformer? Briefly explain them. (03 Marks)

(c) Briefly discuss the advantages and disadvantages of Auto Transformers over double-wound Transformers. (03 Marks)

(d) A 400 kVA transformer has a full copper loss of 3 kW and an iron loss of 2 kW Determine

(i) The output kVA at which the efficiency of the transformer is a maximum.

(ii) The maximum efficiency, assuming the power factor of the load is 0.8.

(05 Marks)

[Total 20 Marks]

Q2.

(a) What are the factors to be consider while designing a good lighting system? (04 Marks)

(b) List out any **6 benefits** of good architecture lighting design? (03 Marks)

(b) What are the types of lamps used in lighting system? Write down their features with typical application?

(08 Marks)

(d) Explain the term **Illuminance** and **Luminance**, also write the Representing formula for Maintenance Factor (M.F) and Utilization Factor (U.F)? (05 Marks)

[Total 20 Marks]

Q3.

(a) (i) State **three** reasons for considering power factor correction.

(03 Marks)

(ii) What are the methods commonly used for power factor correction? (03 Marks)

(b) (i) What are the factors affecting the electricity tariffs? (03 Marks)

(ii) State Three-part electricity tariff briefly. (02 Marks)

(c) (i) How do the cost benefit analysis carried out in implementing new electrical system or installation? (03 Marks)

(ii) What is the information obtained from load curves in power station?

(03 Marks)

(d) Find the value of the capacitance needed to correct a load of 100 kVar at 0.85 lagging power factor to unity power factor. The load is supplied by a 230 V (rms), 50 Hz line. (03 Marks)

[Total 20 Marks]

Q4.

(a) (i) How are three phase transformers made? (02 Marks)

(ii) Why do we need for vector grouping of three phase transformers? (02 Marks)

(iii) What are the vector groupings available in three phase transformers? Briefly explain them. (03 Marks)

(b) (i) State two general types of polyphase induction motors and briefly discuss the differences between them. (03 Marks)

(ii) Draw per phase equivalent circuit of three phase induction motor referred to the stator with indicating the load. (03 Marks)

(c) A 3 phase, 400 V, 50 hp, 50 Hz, four-pole induction machine delivers rated output power at a slip of 0.04 (1hp is 0.746 kW). Determine the,

(i) synchronous speed and motor speed

(ii) speed of the rotating air gap field

(iii) frequency of the rotor circuit

(iv) slip speed in r.p.m

(v) speed of the rotor field relative to the

I. rotor structure

II. stator structure

III. stator rotating field

(07 Marks)

[Total 20 Marks]

Q5.

(a) Explain preliminary energy audit & its importance? (04 Marks)

(b) Explain details energy audit with its 10 steps? (06 Marks)

(c) What is the important of Analysis and Recommendation for Energy Audit? (06 Marks)

(d) Write the short notes on Demand side Management & Write various principles of Energy Management? (04 Marks)

[Total 20 Marks]

Q6.

(a) Differentiate between 'contract demand' and 'maximum demand'? (04 Marks)

(b) What are the types of tariff? Explain the importance of TOD (time of the day) tariff? (05 Marks)

(c) Define the term 'Load Factor' with an example? (03 Marks)

(d) A 250 W sodium vapor lamp is installed on a street. The supply voltage for a street light is 230 V and it operates for around 12 hours in a day. Considering the current of 2 amps and power factor 0.85 calculate the energy consumption per day?

(04 Marks)

(e) A 75 kW three-phase induction motor operates at 55 kW. The measured voltage is 415 V, current is 80 A. Calculate the power factor of the motor?

(04 Marks)

[Total 20 Marks]